CSE7101- Capstone Project
Review-2

PSCS_35: 360-Degree Feedback Software for News Stories in Regional Media (using AI/ML)

Batch Number: CSE_150

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Abstract

This project introduces an AI/ML-driven system to analyze regional news stories about government policies and initiatives. The software identifies sentiment (positive, neutral, negative) and classifies articles into relevant departments using NLP techniques. An interactive dashboard presents the results, giving policymakers, media houses, and researchers a comprehensive 360-degree view of public opinion.

- Automates sentiment analysis, reducing manual bias.
- Provides quick and scalable insights from regional news.
- Facilitates better decision-making and policy communication.
- Lays the foundation for future multilingual and real-time analysis.

IEEE Papers Literature Survey (Paper-1)

Title: Deciphering Political Entity Sentiment in News with LLMs (2024)

What they did:

- Analyzed sentiment toward political leaders and parties in news articles.
- Focused on entity-level sentiment rather than overall text.
- Demonstrated AI replacing manual political news reviews.

How they did it:

- Used large language models (LLMs) for sentiment classification.
- Extracted entities (politicians, departments) before sentiment tagging.
- Validated results against annotated datasets.

Why it matters:

- Shows AI can capture fine-grained sentiment about government entities.
- Similar techniques can be adapted for your project's department classification.
- Provides evidence of AI-driven sentiment outperforming manual analysis.

IEEE Papers Literature Survey (Paper-2)

Title: Prediction of Stock Value Changes Using Sentiment (2024)

What they did:

- Studied how news sentiment influences financial markets.
- Modeled the relationship between sentiment polarity and stock fluctuations.
- Showed news is a key driver of economic signals.

How they did it:

- Extracted sentiment features from online news & web data.
- Applied statistical learning + machine learning models.
- Used real-time stock data for evaluation.

Why it matters:

- Demonstrates sentiment analysis has measurable real-world impact.
- Shows how sentiment can guide decision-making dashboards.
- Strengthens the case for using sentiment-driven feedback in your system.

IEEE Papers Literature Survey (Paper-3)

Title: Fake News Stance Detection Using Deep Learning (2023)

What they did:

- Built models to detect stance (support, oppose, neutral) in news.
- Extended beyond “positive/negative” sentiment.
- Tackled the challenge of fake news credibility.

How they did it:

- Collected fake/real news datasets with stance labels.
- Used deep learning classifiers for multi-class stance detection.
- Trained models on news claims vs. content.

Why it matters:

- Complements your system’s sentiment analysis with stance detection.
- Suggests adding credibility checks to improve reliability.
- Highlights the importance of news authenticity in public perception.

IEEE Papers Literature Survey (Paper-4)

Title: Deep Sentiment Analysis: Case Study on Hausa (2024)

What they did:

- Applied sentiment analysis in Hausa, a low-resource language.
- Explored preprocessing challenges in regional language text.
- Demonstrated scalability of NLP to less-supported languages.

How they did it:

- Used stemming and tokenization for Hausa text cleaning.
- Applied lexicon-based methods for sentiment classification.
- Evaluated performance using manually labeled datasets.

Why it matters:

- Provides strategies for handling Indian regional languages.
- Validates the importance of preprocessing for accuracy.
- Inspires expanding your system beyond English.

IEEE Papers Literature Survey (Paper-5)

Title: Multilingual Sentiment Analysis (2024)

What they did:

- Reviewed multilingual sentiment analysis research.
- Compared rule-based, ML, and deep learning models.
- Explored challenges in cross-language sentiment transfer.

How they did it:

- Collected multilingual datasets.
- Analyzed translation-based vs. native-language approaches.
- Benchmarked models across several languages.

Why it matters:

- Helps plan multilingual expansion of your project.
- Informs model selection for non-English regional news.
- Highlights translation challenges you need to prepare for.

IEEE Papers Literature Survey (Paper-6)

Title: Enhancing Multilingual Hate Speech Detection (2024)

What they did:

- Built models to detect hate speech in multiple languages.
- Tackled cultural/linguistic variations in text meaning.
- Extended classification beyond sentiment polarity.

How they did it:

- Used embeddings trained on multilingual corpora.
- Applied deep learning for contextual classification.
- Benchmarked results against baseline models.

Why it matters:

- Provides strategies for cross-language classification.
- Useful for adapting your system to regional linguistic diversity.
- Shows how context-aware embeddings improve accuracy.

IEEE Papers Literature Survey (Paper-7)

Title: Sentiment Analysis of Tweets Using Emojis and Text (2024)

What they did:

- Studied combined use of emojis + text for sentiment.
- Showed emojis carry strong emotional signals.
- Improved classification accuracy compared to text-only.

How they did it:

- Built datasets of tweets containing both emojis and text.
- Trained multimodal sentiment classifiers.
- Evaluated with precision/recall metrics.

Why it matters:

- Suggests exploring multi-modal cues in your system.
- Inspires use of metadata or extra signals from news sources.
- Shows value of going beyond plain text analysis.

IEEE Papers Literature Survey (Paper-8)

Title: Financial Sentiment Analysis with LLMs & FinBERT (2024)

What they did:

- Applied domain-specific models for financial news analysis.
- Compared generic vs. specialized models for accuracy.
- Focused on fine-grained sentiment categories.

How they did it:

- Used FinBERT pretrained on finance-related corpora.
- Fine-tuned models on labeled financial reports/news.
- Validated performance against human annotations.

Why it matters:

- Shows benefits of specialized models for niche domains.
- Suggests exploring domain-specific training for government news.
- Proves advanced models outperform generic NLP in targeted tasks.

IEEE Papers Literature Survey (Paper-9)

Title: Stock Prediction using Sentiment from Headlines (2024)

What they did:

- Used headlines alone for sentiment-driven stock prediction.
- Proved short text can yield strong predictive signals.
- Linked news events to financial impacts.

How they did it:

- Collected financial headlines datasets.
- Applied sentiment classification + DL predictors.
- Evaluated predictions vs. actual stock changes.

Why it matters:

- Supports analyzing short regional headlines in your system.
- Justifies focusing on quick news snippets for real-time feedback.
- Strengthens link between news sentiment and public outcomes.

IEEE Papers Literature Survey (Paper-10)

Title: Ensemble Learning for Multi-Modal Fake News Detection (2024)

What they did:

- Built hybrid models for fake news detection.
- Combined text, images, and metadata for reliability.
- Outperformed single-model approaches.

How they did it:

- Integrated multiple classifiers in ensemble pipelines.
- Used datasets with multimodal news features.
- Compared against standalone deep learning models.

Why it matters:

- Suggests your project could integrate multiple signals.
- Provides robustness ideas for future enhancement.
- Shows ensemble learning improves accuracy in complex tasks.

Objectives

Main Goal: Develop an AI/ML-based system for analyzing regional news sentiment and departmental classification.

Specific Objectives:

- Automate sentiment analysis of news (Positive, Negative, Neutral).
- Enable automatic departmental classification of news content (Health, Education, Defense, etc.).
- Build a user-friendly dashboard to visualize sentiment trends and department mapping.
- Provide timely, unbiased feedback to policymakers and media houses.
- Lay foundation for future multilingual support and real-time analysis.

Existing Methodology and Drawbacks

Current Challenges

Existing Approaches:

- Manual news review and categorization by analysts.
- Basic sentiment tools (e.g., lexicon-based word counts).
- General-purpose NLP applications not tuned for regional/government context.

Drawbacks:

- Time-consuming & costly – cannot scale with daily news volumes.
- Human bias – inconsistent interpretations.
- Delayed feedback – insights arrive too late for effective decisions.
- Limited scope – little or no department-level categorization.

Proposed Methods and Feasibility Study

Proposed Methods:

- Use TextBlob/ML models for sentiment classification.
- Apply keyword-based and ML-driven rules for department classification.
- Develop a Flask web app with dashboard (Chart.js) for visualization.
- Store results in SQLite database for persistence and historical analysis.

Drawbacks:

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- Human bias – inconsistent interpretations.
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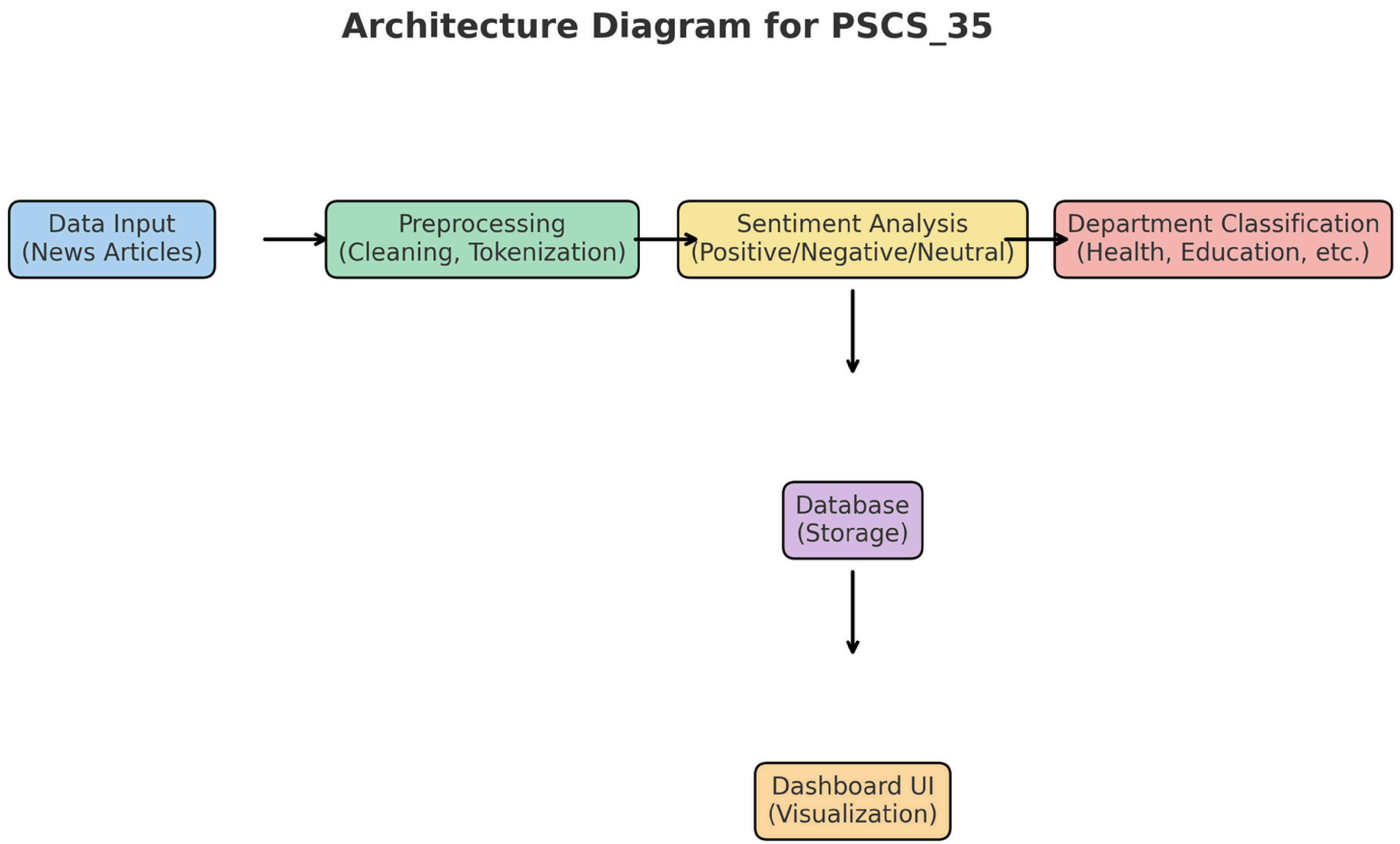
Architecture Diagram



Modules

1. Data Collection Module

- Input from CSV, scraping, or manual entry.

2. Preprocessing Module

- Cleaning, tokenization, stop-word removal.

3. Sentiment Analysis Module

- NLP model determines polarity (Positive/Negative/Neutral).

4. Department Classification Module

- Keyword + ML rules for mapping to departments.

5. Dashboard & Visualization Module

- Flask web app + Chart.js graphs for results.

6. Storage Module (Optional)

- SQLite for persistence & historical analytics.

Hardware and Software Requirements

Hardware (Mac Mini or equivalent):

- Processor: Intel i3 / Apple M1 or higher
- RAM: 8 GB minimum
- Storage: 50 GB free
- Internet: Required for scraping & deployment

Software:

- OS: macOS / Linux / Windows
- Language: Python 3.x
- Libraries: Flask, TextBlob, Pandas, BeautifulSoup4, Chart.js
- Database: SQLite (lightweight, built-in)
- IDE/Editor: VS Code, PyCharm, or Replit online

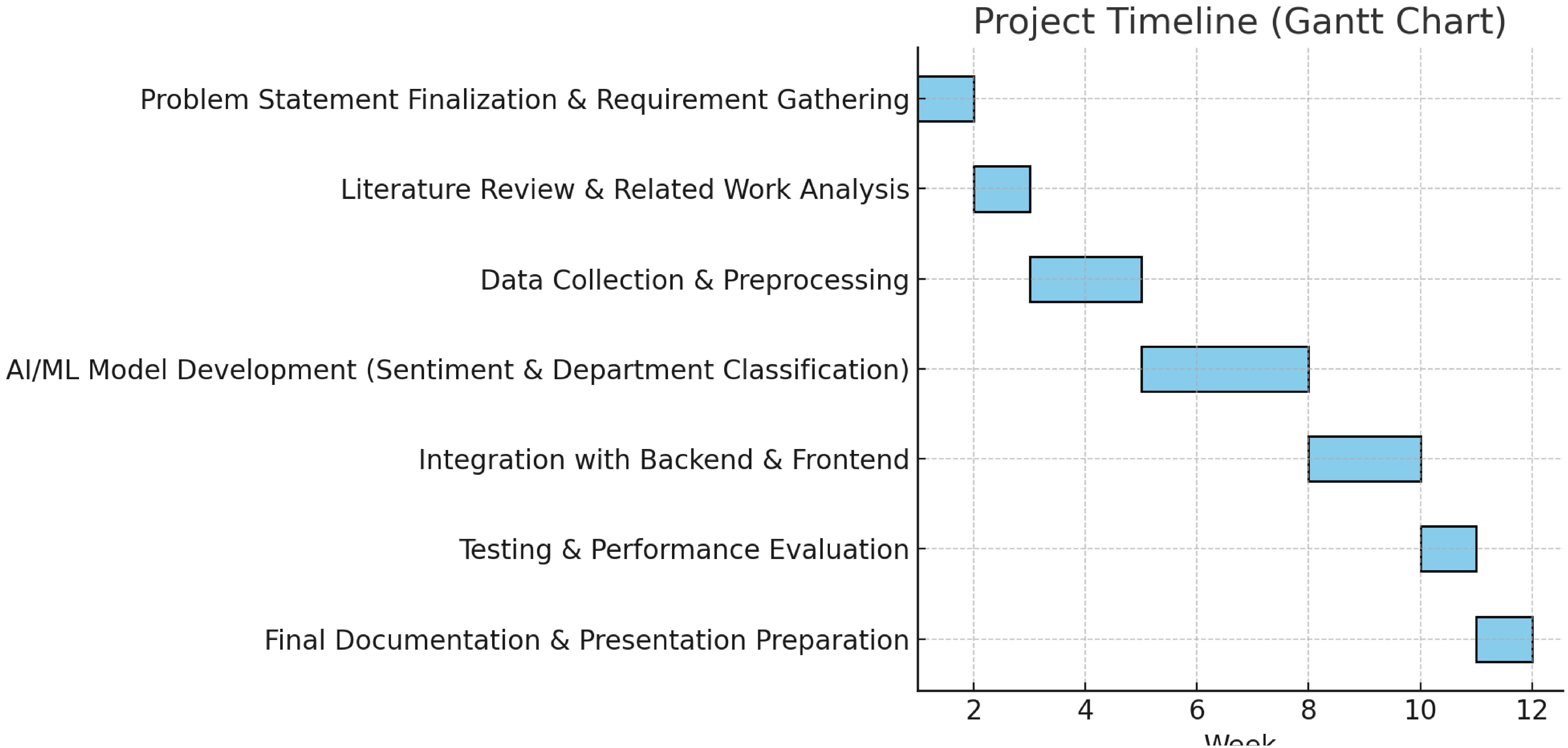
Github Link

Github Link

https://github.com/YashasRaj20231CSE3019/capstone-project-

CSE_150.git

Timeline of the Project (Gantt Chart)



Timeline of the Project (Gantt Chart)

Phases	Task	Duration	Dates
Phase 1	Problem Statement Finalization & Requirement Gathering	1 week	Week 1
Phase 2	Literature Review & Related Work Analysis	1 week	Week 2
Phase 3	Data Collection & Preprocessing	2 weeks	Weeks 3–4
Phase 4	AI/ML Model Development (Sentiment & Department Classification)	3 weeks	Weeks 5–7
Phase 5	Integration with Backend & Frontend	2 weeks	Weeks 8–9
Phase 6	Testing & Performance Evaluation	1 week	Week 10
Phase 7	Final Documentation & Presentation Preparation	1 week	Week 11

References (IEEE Paper format)

All sources referenced in this presentation adhere strictly to the IEEE citation style.

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Thank You!